

Applied Machine Learning

Unit 05 — Lecture 08 Notes

Regression Foundations and Least Squares

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Read this first

These notes are written so that you can learn the material *without* having been in the lecture. Every concept is developed from scratch, every number you see was produced by running the code shown beside it, and every figure was generated from real computation rather than drawn by hand.

Unit II told you *what* to minimise: the mean squared error. Unit III told you *how* to minimise it: take a step downhill and repeat. Unit IV made the table fit to use. This lecture takes the simplest possible model — a straight line — puts the same squared-error objective on it, and asks a question that has not been asked yet in this course: **do we actually have to iterate?**

For this one model the answer is no. Two partial derivatives set to zero give two linear equations in two unknowns, and you solve them once, exactly. The result,

$$b_1 = \frac{S_{xy}}{S_{xx}}, \quad b_0 = \bar{y} - b_1 \bar{x},$$

is the **least-squares estimator**, and deriving it is the examinable core of this lecture. Section 6 does it with every step written out. Do not skim it — write it out yourself on a blank page until you can produce it without looking.

The single result to carry away: on the same five points, gradient descent took **619 iterations** to reproduce what one division produced exactly. On 442 real patients, full-batch gradient descent reproduced $(X^T X)^{-1} X^T y$ to fifteen significant figures. They are the same objective, solved two ways.

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1 How to use these notes

Read in order. Section 6 is the spine of the lecture, and Sections 8 to 11 are all consequences of it. Everything here assumes only what Unit III assumed: partial derivatives, a 2×2 matrix inverse, and the idea of a loss function. The one new piece of mathematics is setting a gradient to zero and *solving* rather than stepping.

There are four kinds of box. **Key idea** — the one sentence to remember a month from now. **Worked example** — a calculation done by hand, with small numbers, before any library is used. **Caution** — the specific mistake that section exists to prevent. **Try it yourself** — something to run, with the expected output given so you can check.

Code appears in a framed block, and directly beneath it a shaded block showing what that code *actually printed*. If you run it and get something different, trust your machine and tell me.

1.1 What you need

```
pip install numpy scipy scikit-learn matplotlib
```

Every number in these notes was produced with:

```
import numpy, scipy, sklearn, matplotlib
print("numpy", numpy.__version__, "| scipy", scipy.__version__,
      "| sklearn", sklearn.__version__,
      "| matplotlib", matplotlib.__version__)
```

```
numpy 2.2.6 | scipy 1.15.3 | sklearn 1.7.2 | matplotlib 3.10.9
```

One bundled dataset carries the real-data half of the lecture, and it **does not download anything**.

```
from sklearn.datasets import load_breast_cancer, load_diabetes, load_wine
dia = load_diabetes(); Xd, yd = dia.data, dia.target.astype(float)
bc = load_breast_cancer()
wine = load_wine()
```

```
load_diabetes      442 rows x 10 features, continuous target
  features: age, sex, bmi, bp, s1, s2, s3, s4, s5, s6
  target range 25.0 to 346.0 (disease progression one year on)
load_breast_cancer 569 rows x 30 features
load_wine          178 rows x 13 features
missing cells anywhere: 0
```

Do not reach for fetch_*

Every `fetch_*` loader downloads over HTTPS, and on the campus network those downloads fail with a certificate error you cannot fix from your machine. Nothing in this course depends on one: use the bundled loaders (`load_diabetes`, `load_breast_cancer`, `load_wine`, `load_iris`, `load_digits`) or a fixed-seed synthetic set.

One seed — the convention for this lecture

Everything in these notes is computed from the datasets above — bundled `load_diabetes`, the published Anscombe table, and the five-point revision-and-quiz example — together with **one seed**. That seed is **0**: every random number generator is created as `np.random.default_rng(0)`, every shuffled split is made with `random_state=0`, and every listing that draws, shuffles or samples shows that line rather than hiding it in a helper. None of the datasets in this lecture is itself random — they are either bundled or written out

by hand — so there is nothing else to fix. Every number printed in these notes therefore reproduces *exactly* if you run the code as written. If you get something different, the difference is in your code or your library versions, not in chance — which is the only thing that makes a printed number worth anything to you.

1.2 Abbreviations and notation

Each abbreviation is also spelled out where it first appears in the text.

Short form	Meaning
SLR	simple linear regression: one predictor, a straight line
SSE	sum of squared errors, $\sum_i (y_i - \hat{y}_i)^2 = \sum_i e_i^2$
SST	total sum of squares, $\sum_i (y_i - \bar{y})^2$
SSR	regression (explained) sum of squares, $\sum_i (\hat{y}_i - \bar{y})^2$
SAE	sum of absolute errors, $\sum_i e_i $
LAD	least absolute deviations: the line that minimises SAE
RMSE	root mean squared error, $\sqrt{\text{SSE}/n}$, in the units of y
GD, SGD	gradient descent; stochastic gradient descent
SVD	singular value decomposition (a matrix factorisation)
sd	standard deviation
BMI	body mass index
k -NN	k -nearest neighbours
L3, L9, ...	Lecture 3, Lecture 9, ... of this course
OLS	ordinary least squares — the method of this lecture
ANOVA	analysis of variance (the SST = SSR + SSE split, Lecture 9)
CO1, CO2, ...	course outcomes: the official outcomes of this course that each learning outcome below serves
$\bar{x}, \bar{y}$	means of x and y (<code>xbar</code> , <code>ybar</code> in printed output)
$\hat{y}_i$	fitted value $b_0 + b_1 x_i$ (<code>yhat</code>)
e_i	residual $y_i - \hat{y}_i$
S_{xx}, S_{xy}, S_{yy}	$\sum (x_i - \bar{x})^2, \sum (x_i - \bar{x})(y_i - \bar{y}), \sum (y_i - \bar{y})^2$ (<code>Sxx</code> , ...)
$X^\top$	transpose of X (<code>X'</code> or <code>X.T</code> in code)

1.3 Learning outcomes

By the end of this lecture you should be able to (the tag in brackets is the course outcome, CO, that each item serves):

1. Define a regression problem, distinguish it from classification, name its components (y, x, ε, β), and give worked examples of regression from at least three application areas (CO1).
2. Classify a stated model as linear or nonlinear *in the parameters*, and say which of the standard regression models are fitted by solving a linear system and which need an iterative solver (CO1).
3. List and apply the steps in a regression analysis — problem, variables, data, specification, fitting method, fit, validation, use — end to end on a real dataset (CO1, CO2).
4. **Derive the least-squares estimator for simple linear regression:** form the sum of squared errors, take both partial derivatives, set them to zero, obtain the normal equations, solve them for $b_1 = S_{xy}/S_{xx}$ and $b_0 = \bar{y} - b_1 \bar{x}$, and verify the second-order condition (CO3).
5. Carry out that computation numerically on a small table by hand, and confirm it against `np.polyfit` and `sklearn.linear_model.LinearRegression` (CO2, CO3).

- Write the normal equations in matrix form, invert a 2×2 $X^T X$ by hand, and state the rank condition under which $(X^T X)^{-1} X^T y$ exists (CO3).
- Prove and use the properties of the line of best fit: residuals sum to zero with an intercept, the line passes through $(\bar{x}, \bar{y})$, residuals are orthogonal to the predictor, and $SST = SSR + SSE$ (CO3).
- Explain the relationship between the closed-form solution and the gradient descent of Lecture 3 — same objective, two routes — and say which to use when (CO2).
- Identify what least squares cannot tell you: leverage, the influence of a single point, and the fact that four datasets can share every computed statistic (CO2, CO3).

2 What regression is

Regression is the modelling of a quantitative response as a function of one or more predictors. The response y is a number on a continuous scale — a price in rupees, a temperature, a mark, a disease-progression score. If y were a label from a finite set (spam / not spam, benign / malignant) the problem would be classification, which is Unit VI.

The model always has the same two-part shape:

$$y = \underbrace{f(x)}_{\text{systematic}} + \underbrace{\varepsilon}_{\text{random}}. \quad (1)$$

The systematic part is what the predictors explain. The random part is everything else: measurement error, variables you did not record, genuine irreducible variation. No amount of modelling removes ε ; the job is to get f right and leave ε as small and as structureless as the data permits.

The vocabulary is worth fixing now, because the examination uses all of it.

Symbol	Names you will meet	In our running example
y	response, dependent variable, target, label	quiz score out of 10
x	predictor, regressor, independent variable, feature	hours of revision
ε	error, disturbance, noise term	everything else that affects the mark
β_0, β_1	population parameters, true coefficients	unknown, and always will be
b_0, b_1	estimates, fitted coefficients	2.2 and 0.6, computed from five rows

2.1 Two jobs, and you must choose

A regression can be built for **prediction** — given a new x , what is y ? — or for **explanation** — which predictors matter, in which direction, and by how much? The two jobs are judged differently. Prediction is judged by error on data the model has not seen. Explanation is judged by whether the coefficients can be trusted, which is a much stronger requirement.

Lecture 6 gave the sharpest possible illustration of the difference: with an intercept and all k dummy columns present, two *completely different* coefficient vectors gave identical predictions to 0.00e+00. The model predicted perfectly well and the coefficients meant nothing.

Key idea

Decide whether you are predicting or explaining *before* you choose the model, because the two goals justify different choices and are judged by different evidence. Simple linear regression does both, which is one reason it is still the first model taught.

2.2 Where regression is used

Field	Response y	Predictors x
Medicine	disease progression after a year	BMI, blood pressure, six serum measures
Real estate	sale price, in rupees	area, rooms, age, locality
Energy	tomorrow's peak load	temperature, weekday, last week's load
Agriculture	yield per hectare	rainfall, fertiliser, soil nitrogen
Manufacturing	tensile strength	alloy fractions, furnace temperature
Finance	next-day return	yesterday's return, volume, volatility
Education	end-semester mark	attendance, internal marks, hours studied
Environment	river discharge	rainfall, catchment area, season

The medical row is the one we will actually compute with. `load_diabetes` holds 442 patients, ten baseline measurements each, and a quantitative score of disease progression one year later. It is bundled with scikit-learn, so it is available on a machine with no network at all.

3 Regression examples and models

3.1 The family

Model	Form	How it is fitted
Simple linear	$y = \beta_0 + \beta_1 x + \varepsilon$	least squares, closed form
Multiple linear	$y = \beta_0 + \sum_j \beta_j x_j + \varepsilon$	least squares, closed form (L11)
Polynomial	$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots$	least squares, closed form
Transformed predictors	$y = \beta_0 + \beta_1 \log x + \varepsilon$	least squares, closed form
Ridge / lasso	least squares plus a penalty	closed form / convex solver (L12)
Logistic	$\Pr(y = 1) = \sigma(\beta_0 + \beta_1 x)$	maximum likelihood, iterative (L10)
Nonlinear	$y = \theta_1 e^{-\theta_2 x} + \varepsilon$	numerical optimisation
Nonparametric	regression trees, k -NN, splines	no single global equation

The last column uses three words that recur all lecture. A **closed form** answer is a formula: substitute the data, compute once, and you are done. An **iterative** method (gradient descent, Lecture 3) instead repeats an update until the numbers stop changing. **Convex** means bowl-shaped; it is defined properly in Section 5.2.

3.2 “Linear” means linear in the parameters

This is the definition students most often get wrong, and it is a favourite examination question.

Key idea

A regression model is **linear** when it is linear in the *parameters*, not in the predictors. $y = \beta_0 + \beta_1 x + \beta_2 x^2$ draws a parabola and is a linear model. $y = \beta_0 + \beta_0 \beta_1 x$ draws a straight line and is *not* a linear model, because the slope is a product of two parameters.

The reason this matters is entirely practical: a model that is linear in the parameters can be written as $y = Xb + e$ for some design matrix X , and Section 7 will show that such a model is fitted by solving one linear system. Everything else needs an iterative solver, a starting point, and luck.

One solver, three models

Fit degree 1, 2 and 3 polynomials to the same 40 points, each time by building a design matrix and solving the normal equations directly.

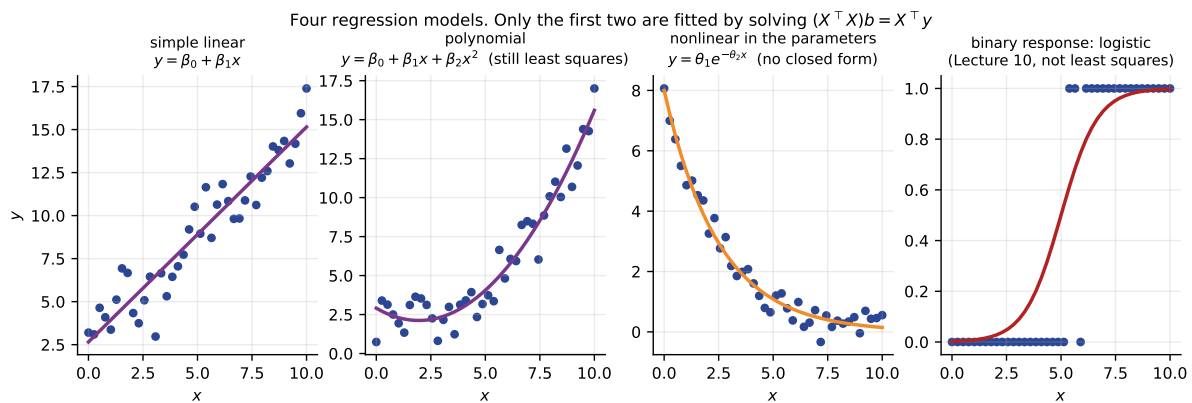


Figure 1: Four regression models. Panels 1 and 2 are both fitted by solving $(X^T X)b = X^T y$; only the columns of X differ. Panel 3 has a parameter inside an exponential, so no closed form exists. Panel 4 has a 0/1 response, so squared error is the wrong objective altogether — that is Lecture 10.

```

xp = np.linspace(-3.0, 3.0, 40)
rng = np.random.default_rng(0) # the one seed for this lecture
yp = 2.0 - 0.5*xp + 0.8*xp**2 + rng.normal(0, 1, 40)
for deg in (1, 2, 3):
    P = np.column_stack([xp**k for k in range(deg + 1)])
    bhat = np.linalg.solve(P.T @ P, P.T @ yp)
    res = yp - P @ bhat
    print(deg, P.shape, np.round(bhat, 4), np.sum(res**2))

```

degree 1	design (40, 2)	coefficients [4.4626 -0.4485]	SSE 265.953
degree 2	design (40, 3)	coefficients [1.7027 -0.4485 0.8751]	SSE 22.655
degree 3	design (40, 4)	coefficients [1.7027 -0.4331 0.8751 -0.0027]	SSE 22.649

The data came from $y = 2 - 0.5x + 0.8x^2 + \varepsilon$, and the degree-2 fit recovers those coefficients to within the noise. **The same three lines of code fitted all three models.** Everything this lecture proves about the straight line transfers unchanged to any design matrix.

4 The steps in a regression analysis

A regression analysis is a workflow, not a function call. The version below follows Chatterjee and Hadi's classical presentation and is the one you should be able to reproduce from memory.

1. **Statement of the problem.** What question are you answering, in words, before any data is touched? “Does body mass index predict disease progression?” is a question. “Run a regression on the diabetes data” is not.
2. **Selection of the variables.** Which quantity is y ? Which quantities could be x ? In what units, on what scale, measured how?
3. **Data collection.** How many rows, gathered how, and split how. This is where the Unit IV material lives: cleaning, encoding, scaling and a train/test split made *before* anything is fitted.
4. **Model specification.** Write the equation down explicitly: $y = \beta_0 + \beta_1 x + \varepsilon$, with the assumptions you are making about ε .
5. **Choice of fitting method.** Least squares, maximum likelihood, penalised least squares, a robust criterion. This lecture is about the first; Lecture 9 shows the first two coincide under Normal (Gaussian) noise. (*Maximum likelihood* picks the parameter values under which the

observed data would have been most probable.)

6. **Fitting the model.** Compute the estimates. On the scale of problems in this course this is the fastest step of the eight.
7. **Model validation and criticism.** Do the residuals look like noise? Is the held-out error close to the training error? Are the assumptions of step 4 defensible? If not, go back to step 4 — not to step 6.
8. **Using the model.** Predict, or interpret — and say which.

The analysis is usually lost before step 6

By the time you type `fit()` every decision that matters has already been made. A perfect fit to the wrong response answers the wrong question perfectly, and no amount of cross-validation will tell you so.

4.1 All eight steps, on real patients

```
j = list(dia.feature_names).index("bmi")
xr = Xd[:, j].reshape(-1, 1)
xtr, xte, ytr, yte = train_test_split(xr, yd, test_size=0.25, random_state=0)
m = LinearRegression().fit(xtr, ytr)
```

```
1 problem : does body mass index predict disease progression?
2 variables : y = progression score; x = bmi (centred and scaled here)
3 data      : 442 patients, 331 train / 111 test
4 model     : y = b0 + b1 x + e
5 method    : least squares
6 fit       : LinearRegression  b0 = 153.2257  b1 = 1016.9235
              by hand          b0 = 153.2257  b1 = 1016.9235
7 check     : train RMSE 61.711  test RMSE 64.664
              training residuals sum to 2.274e-12
              5-fold RMSE 62.605 +/- 2.888
              always predicting ybar gives RMSE 77.006
8 use       : one sd more bmi predicts 49.4 more progression points
```

Three things in that output are worth reading twice. First, line 6: the library and the hand formula of Section 6 agree to four decimal places, which is the whole point of this lecture. Second, line 7: the cross-validated RMSE (root mean squared error, $\sqrt{\text{SSE}/n}$, in the units of y) of 62.6 against 77.0 for the null model that always predicts $\bar{y}$ — *one* column of *one* measurement removes about a fifth of the error. Third, also line 7: the test RMSE (64.7) is close to the training RMSE (61.7), so there is no over-fitting here. With two parameters and 331 rows there was never going to be.

5 The simple linear regression model

5.1 The model and its assumptions

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n. \quad (2)$$

The four standard assumptions are:

1. **Linearity.** $\mathbb{E}[y \mid x] = \beta_0 + \beta_1 x$: the conditional mean of the response really is a straight line in x .
2. **Independence.** The ε_i are independent of one another. Time series and repeated measures violate this routinely.

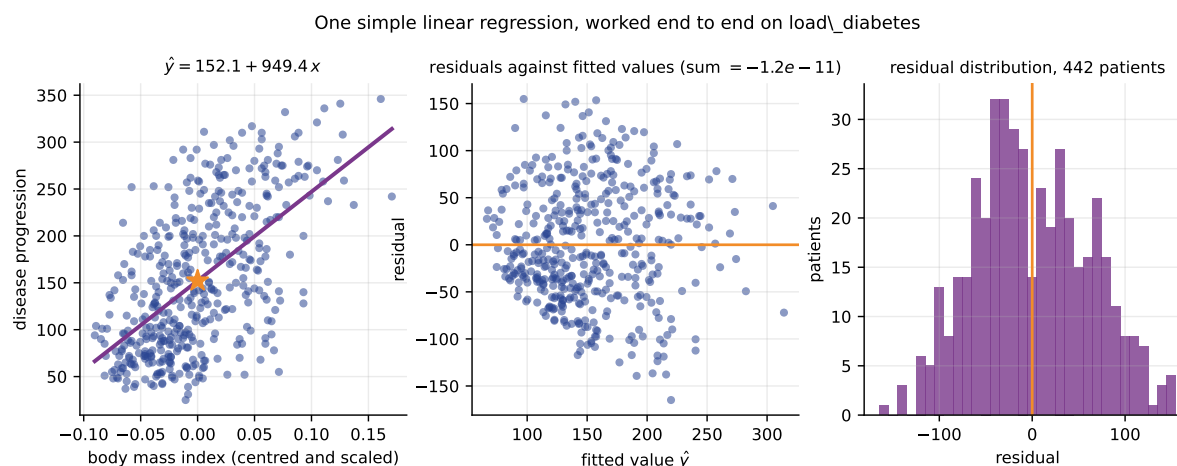


Figure 2: Step 7 on the fit of Section 4.1. Left: the fitted line passes through the orange star at $(\bar{x}, \bar{y})$ — proved in Section 9. Middle: residuals against fitted values, summing to 10^{-11} — also proved there. Right: their distribution. Note that the spread widens towards the right of the middle panel, which is a real defect of this model and exactly what Lecture 9’s diagnostics exist to catch.

3. **Constant variance** (homoscedasticity). $\text{Var}(\varepsilon_i) = \sigma^2$, the same at every x .

4. **Zero mean.** $\mathbb{E}[\varepsilon_i] = 0$, so the errors do not systematically push the response one way.

Notice what is *not* on the list: Normality. You do not need Normal errors to compute the least-squares estimates or to establish any property in Section 9. Normality enters in Lecture 9, when we want confidence intervals, p -values and the maximum-likelihood interpretation.

Greek letters and Roman letters are different things

β_0, β_1 describe the population. They are fixed, unknown, and you will never see them. b_0, b_1 are computed from your n rows; collect the data again and they change. Likewise ε_i is the unobservable error and $e_i = y_i - \hat{y}_i$ is the observable residual. Writing “ $\varepsilon_i = 0.6$ ” in an answer script costs marks and, more importantly, means you have not seen the distinction.

5.2 Four words you need before the derivation

The rest of the lecture leans on four pieces of vocabulary from calculus and optimisation. They are defined here once.

Closed form. The answer can be written as a formula in the data — plug in, compute once. The opposite is an *iterative* method, which improves a guess step by step until it stops changing.

Stationary point. A point where every partial derivative of the function is zero, so the surface is flat there. A stationary point can be a minimum, a maximum, or a *saddle* (a minimum in one direction and a maximum in another; see Section 6.5).

Local and global minimum. A local minimum is lower than every point *near* it; a global minimum is lower than every point *anywhere*. A function can have many local minima and only one global minimum value.

Convex function. f is convex if the straight chord joining any two points of its graph never lies *below* the graph:

$$f(\lambda u + (1 - \lambda)v) \leq \lambda f(u) + (1 - \lambda)f(v) \quad \text{for all } u, v \text{ and } 0 \leq \lambda \leq 1.$$

Picture a bowl. f is *strictly* convex if the inequality is strict whenever $u \neq v$: the bowl then has exactly one bottom. In one variable, $f''(x) > 0$ everywhere guarantees strict convexity (for

example x^2); $x^4 - x^2$ is not convex — it has two separate dips.

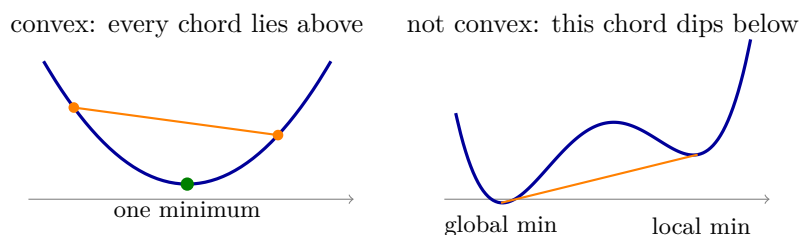


Figure 3: Left: a convex function. Every chord (orange) lies above the graph, and there is one minimum. Right: a function that is not convex. The chord between its two dips passes below the graph, and “the derivative is zero” holds at a local minimum, the global minimum and the hump between them.

Why convexity matters

For a convex function **every stationary point is a global minimum**. So if the least-squares criterion is convex — and Section 6.5 proves it is — then “set the derivatives to zero” finds the best line, not merely a locally good one.

5.3 Which line? Choosing a criterion

Infinitely many straight lines pass near a cloud of points. To pick one we need a criterion: a number, computable from the data and the candidate line, that we then make as small as possible. Four candidates:

- $\sum_i e_i$. **Useless.** Positive and negative errors cancel, so infinitely many lines score exactly zero — including some very bad ones. We will meet three of them in Section 8.3.
- $\sum_i |e_i|$, the sum of absolute errors (SAE); the line that minimises it is the least absolute deviations (LAD) line. Sensible and robust to outliers, and it has *no closed form*: $|e|$ has a corner at $e = 0$, so “differentiate and set to zero” is not available. Its minimiser need not even be unique.
- $\max_i |e_i|$ (minimax, Chebyshev). Lets a single bad row dictate the whole line.
- $\sum_i e_i^2$ (**least squares**). Differentiable everywhere, convex, and — the decisive property — its derivative is *linear* in b_0 and b_1 .

Key idea

Least squares is not chosen because squared error is the truest measure of wrongness. It is chosen because the derivative of a square is linear, so setting the gradient to zero produces a *linear system*, which has a formula. Every other criterion on the list needs a numerical search.

The cost of that convenience is sensitivity: squaring an error of 4 gives 16, so a single distant point dominates the criterion. We measure exactly how much in Section 12.1.

Squares against absolutes on five points

```
least squares          b0 = 2.2000  b1 = 0.6000  SSE = 2.4000  SAE = 3.2000
least absolute deviation b0 = 1.2500  b1 = 0.7500  SSE = 3.8750  SAE = 3.0000
the LAD line passes exactly through points 1 and 5
LAD residuals: [ 0.    1.25  1.5  -0.25  0. ]
a DIFFERENT line    b0 = 1.4340  b1 = 0.7132  SAE = 3.0000
```

Each criterion wins on its own scale, as it must: least squares has the smaller SSE, least

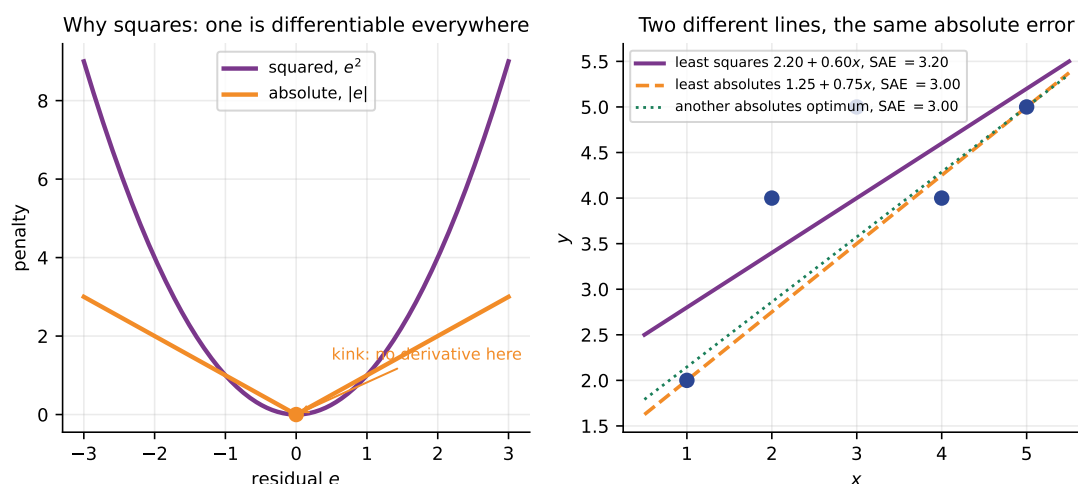


Figure 4: Left: $|e|$ has a corner at the origin, where it has no derivative; e^2 is smooth everywhere. Right: on the five points of Section 8 two different lines achieve *exactly* the same total absolute error of 3.00, so the least-absolute-deviation answer is not unique. Least squares always has exactly one answer.

absolute deviations the smaller SAE. The last line is the point of the box — **two different lines, the same absolute error**.

6 Least-squares estimation: the derivation

This section is the examinable core of the lecture. Work through it with a pen.

6.1 The object being minimised

For a candidate line $\hat{y} = b_0 + b_1x$ the residual at row i is $e_i = y_i - b_0 - b_1x_i$, and the criterion is the **sum of squared errors**

$$\text{SSE}(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1x_i)^2. \quad (3)$$

Two observations before differentiating, both of which students skip and then regret.

1. In (3) the x_i and y_i are **known numbers**. The variables are b_0 and b_1 — two of them, whatever n is. SSE is a function of two real variables and nothing else.
2. Every term is a square, so $\text{SSE} \geq 0$ and the surface is a bowl. It is bounded below and, as we will confirm in Section 6.5, strictly convex. It therefore has exactly one lowest point.

This is the Lecture 3 objective

Lecture 3 minimised $J(w) = \frac{1}{n} \|Xw - y\|^2$, which is precisely $\frac{1}{n} \text{SSE}$. Dividing a function by a positive constant does not move its minimiser. **Same function, same answer — and this lecture finds the answer without stepping.**

6.2 First-order conditions

At a minimum both partial derivatives vanish. Differentiate (3) term by term, using the chain rule on $(y_i - b_0 - b_1x_i)^2$. The inner derivative with respect to b_0 is -1 , and with respect to b_1

is $-x_i$:

$$\frac{\partial \text{SSE}}{\partial b_0} = \sum_{i=1}^n 2(y_i - b_0 - b_1 x_i) \cdot (-1) = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i), \quad (4)$$

$$\frac{\partial \text{SSE}}{\partial b_1} = \sum_{i=1}^n 2(y_i - b_0 - b_1 x_i) \cdot (-x_i) = -2 \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i). \quad (5)$$

The bracket in both is the residual e_i . So the two derivatives are $-2 \sum_i e_i$ and $-2 \sum_i x_i e_i$ — remember that, because setting them to zero is going to hand us two properties of the fitted line for nothing (Section 9).

6.3 The normal equations

Set (4) and (5) to zero and divide through by -2 :

$$\sum_i y_i - n b_0 - b_1 \sum_i x_i = 0 \quad \implies \quad n b_0 + b_1 \sum_i x_i = \sum_i y_i, \quad (6)$$

$$\sum_i x_i y_i - b_0 \sum_i x_i - b_1 \sum_i x_i^2 = 0 \quad \implies \quad b_0 \sum_i x_i + b_1 \sum_i x_i^2 = \sum_i x_i y_i. \quad (7)$$

(The middle step uses $\sum_i b_0 = n b_0$, since b_0 does not depend on i .)

Equations (6) and (7) are the **normal equations**. Look at what has happened: a minimisation over a continuum of lines has turned into *two linear equations in two unknowns*, whose coefficients — n , $\sum x_i$, $\sum x_i^2$, $\sum y_i$, $\sum x_i y_i$ — are five numbers you can accumulate in a single pass over the data.

6.4 Solving them

Divide (6) by n and use $\frac{1}{n} \sum_i x_i = \bar{x}$, $\frac{1}{n} \sum_i y_i = \bar{y}$:

$$b_0 + b_1 \bar{x} = \bar{y} \quad \implies \quad b_0 = \bar{y} - b_1 \bar{x}. \quad (8)$$

Substitute (8) into (7):

$$(\bar{y} - b_1 \bar{x}) \sum_i x_i + b_1 \sum_i x_i^2 = \sum_i x_i y_i,$$

and collect the b_1 terms on one side:

$$b_1 \left(\sum_i x_i^2 - \bar{x} \sum_i x_i \right) = \sum_i x_i y_i - \bar{y} \sum_i x_i.$$

Now use $\sum_i x_i = n \bar{x}$ on both sides:

$$b_1 \left(\sum_i x_i^2 - n \bar{x}^2 \right) = \sum_i x_i y_i - n \bar{x} \bar{y}. \quad (9)$$

The two brackets in (9) have names. Define

$$S_{xx} = \sum_i (x_i - \bar{x})^2, \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}), \quad S_{yy} = \sum_i (y_i - \bar{y})^2. \quad (10)$$

The identification is a two-line expansion, and it is worth doing once because it recurs in every regression proof you will meet:

$$\begin{aligned} \sum_i (x_i - \bar{x})^2 &= \sum_i x_i^2 - 2\bar{x} \sum_i x_i + \sum_i \bar{x}^2 = \sum_i x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum_i x_i^2 - n\bar{x}^2, \\ \sum_i (x_i - \bar{x})(y_i - \bar{y}) &= \sum_i x_i y_i - \bar{y} \sum_i x_i - \bar{x} \sum_i y_i + n\bar{x}\bar{y} \\ &= \sum_i x_i y_i - n\bar{x}\bar{y} - n\bar{x}\bar{y} + n\bar{x}\bar{y} = \sum_i x_i y_i - n\bar{x}\bar{y}. \end{aligned}$$

So (9) reads $b_1 S_{xx} = S_{xy}$, and provided $S_{xx} \neq 0$:

The least-squares estimator

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad b_0 = \bar{y} - b_1 \bar{x}.$$

Two remarks. First, the pair of forms in (10) — deviations on the left, raw sums on the right — are numerically different: the deviation form is more accurate, the raw-sum form is what you use in an examination because $\sum x_i$, $\sum x_i^2$ and $\sum x_i y_i$ come straight off a five-row table. Second, dividing S_{xy} and S_{xx} by $n - 1$ turns them into the sample covariance and variance, so

$$b_1 = \frac{\text{cov}(x, y)}{\text{var}(x)} = r_{xy} \frac{s_y}{s_x}, \quad (11)$$

where r_{xy} is the Pearson correlation. The slope is the correlation, rescaled by the ratio of the two standard deviations.

6.5 Second-order condition: it really is a minimum

A vanishing gradient marks a minimum, a maximum or a saddle. In one variable you would settle the question with the second-derivative test: $f'(x) = 0$ and $f''(x) > 0$ means a minimum. With several variables the second derivative becomes a matrix.

Hessian. The matrix H of all second partial derivatives, with entry (j, k) equal to $\partial^2 f / \partial b_j \partial b_k$. It is symmetric, and it records how the surface curves in every direction.

Positive definite. A symmetric matrix H is positive definite if $v^\top H v > 0$ for every non-zero vector v . Geometrically: whichever direction v you walk from the stationary point, the surface curves *upward*. For a 2×2 matrix it is enough to check that $H_{11} > 0$ and $\det H > 0$. Equivalently, all the *eigenvalues* of H are positive; the eigenvalues are the curvatures along the principal axes of the bowl.

Saddle. A stationary point where the surface curves up in some directions and down in others. The standard example is $f(u, v) = u^2 - v^2$ at the origin: $H = \text{diag}(2, -2)$ has one positive and one negative eigenvalue, so it is not positive definite.

The multivariable test is then: gradient zero and Hessian positive definite $\Rightarrow$ strict local minimum. If the Hessian is positive definite *everywhere*, the function is strictly convex and that minimum is the unique global one (Section 5.2).

Differentiate (4) and (5) again:

$$\frac{\partial^2 \text{SSE}}{\partial b_0^2} = 2n, \quad \frac{\partial^2 \text{SSE}}{\partial b_1^2} = 2 \sum_i x_i^2, \quad \frac{\partial^2 \text{SSE}}{\partial b_0 \partial b_1} = 2 \sum_i x_i.$$

The Hessian is therefore

$$H = 2 \begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix}, \quad \det H = 4 \left(n \sum_i x_i^2 - (\sum_i x_i)^2 \right). \quad (12)$$

Now $n \sum_i x_i^2 - (\sum_i x_i)^2 = n \sum_i x_i^2 - n^2 \bar{x}^2 = n(\sum_i x_i^2 - n \bar{x}^2) = n S_{xx}$, so

$$\det H = 4n S_{xx}. \quad (13)$$

H is a symmetric 2×2 matrix. It is positive definite exactly when its leading entry and its determinant are both positive. The leading entry is $2n > 0$; the determinant is $4n S_{xx} > 0$ whenever the x_i are not all equal. So:

Second-order condition

H is positive definite, SSE is **strictly convex**, and the stationary point found above is the **unique global minimum**. There is one least-squares line and no other.

The exception is instructive. If every x_i takes the same value then $S_{xx} = 0$, the data is a vertical strip, and no line through it is determined — infinitely many lines give the same SSE. The algebra tells you so by refusing to divide.

The second-order condition, numerically

```
Xm = np.column_stack([np.ones(n), x])
H = 2.0 * Xm.T @ Xm # Hessian of SSE(b0, b1)
print(H, np.linalg.det(H), np.linalg.eigvalsh(H))
```

```
Hessian of SSE =
[[ 10.  30.]
 [ 30. 110.]]
d2SSE/db0^2 = 2n      = 10.0
determinant      = 200.0
and 4*n*Sxx      = 200.0
eigenvalues      = [ 1.6905 118.3095]
positive definite -> the stationary point is a strict minimum: True
if every x is identical: Sxx = 0.0 and det(X'X) = 0.0 -> no unique fit
```

$\det H = 200$ and $4nS_{xx} = 4 \cdot 5 \cdot 10 = 200$, exactly as (13) says. Both eigenvalues are positive, so the bowl curves upward in every direction.

7 The matrix form

Everything above generalises, and the generalisation is shorter than the special case.

Stack the rows. Let X be the $n \times 2$ **design matrix** whose first column is all ones and whose second column is x ; let y be the $n \times 1$ response and $b = (b_0, b_1)^\top$. Then $\hat{y} = Xb$ and

$$\text{SSE}(b) = \|y - Xb\|^2 = (y - Xb)^\top (y - Xb) = y^\top y - 2b^\top X^\top y + b^\top X^\top X b. \quad (14)$$

(The two cross terms $b^\top X^\top y$ and $y^\top X b$ are equal because each is a 1×1 matrix and each is the transpose of the other.) Using $\partial(a^\top b)/\partial b = a$ and $\partial(b^\top A b)/\partial b = 2Ab$ for symmetric A :

$$\frac{\partial \text{SSE}}{\partial b} = -2X^\top y + 2X^\top X b = 0 \quad \implies \quad \boxed{X^\top X b = X^\top y} \quad (15)$$

and, when $X^\top X$ is invertible,

$$b = (X^\top X)^{-1} X^\top y. \quad (16)$$

The Hessian is $2X^\top X$, which is positive definite exactly when X has **full column rank**, meaning no column of X is a linear combination of the others — the general form of the “not all x_i equal” condition (a constant x column would be a multiple of the column of ones). Note that (15) never mentions the number of columns: the same three lines fit a multiple regression with fifty predictors, which is Lecture 11 already done.

The matrix route on five points, entirely by hand

With $x = (1, 2, 3, 4, 5)$ and $y = (2, 4, 5, 4, 5)$:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{pmatrix}, \quad X^T X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \quad X^T y = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 20 \\ 66 \end{pmatrix}.$$

Here $\sum x_i = 15$, $\sum x_i^2 = 1+4+9+16+25 = 55$, $\sum y_i = 20$ and $\sum x_i y_i = 2+8+15+16+25 = 66$. Then

$$\det(X^T X) = 5 \cdot 55 - 15^2 = 275 - 225 = 50 = n S_{xx}, \quad (X^T X)^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix},$$

$$b = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} \begin{pmatrix} 20 \\ 66 \end{pmatrix} = \frac{1}{50} \begin{pmatrix} 1100 - 990 \\ -300 + 330 \end{pmatrix} = \frac{1}{50} \begin{pmatrix} 110 \\ 30 \end{pmatrix} = \begin{pmatrix} 2.2 \\ 0.6 \end{pmatrix}.$$

The same answer as the scalar formulas, by a completely different route.

```
Xm = np.column_stack([np.ones(n), x])
beta = np.linalg.inv(Xm.T @ Xm) @ (Xm.T @ y)
```

```
X'X =
[[ 5. 15.]
 [15. 55.]]
X'y = [20. 66.]
det(X'X) = 50.0000    and n*Sxx = 50.0000
inv(X'X) =
[[ 1.1 -0.3]
 [-0.3  0.1]]
beta = inv(X'X) X'y = [2.2 0.6]
np.linalg.solve      = [2.2 0.6]
np.linalg.lstsq      = [2.2 0.6]
np.linalg.pinv       = [2.2 0.6]
agrees with the scalar formulas to 2.665e-15
```

The printed inverse is the fraction from the board: $55/50 = 1.1$ and $-15/50 = -0.3$.

Never write `np.linalg.inv` in real code

It gives the same answer here, and worse answers than `np.linalg.solve` or `np.linalg.lstsq` whenever $X^T X$ is close to singular (almost non-invertible: its determinant is near zero, so the inverse is huge and unreliable) — which, as Lecture 6 showed with the dummy variable trap (condition number 1.2×10^{15}), is not a rare event. Use `solve` when you know the matrix is well conditioned and `lstsq` when you do not. `LinearRegression` uses `lstsq` internally.

8 The five-point calculation, by hand and in code

8.1 By hand

Five students. x is hours of revision in the week before a quiz; y is the quiz score out of ten. The numbers are chosen so that every intermediate quantity is an integer or a clean decimal.

Least squares on five points

						Σ
x	1	2	3	4	5	15
y	2	4	5	4	5	20
$x - \bar{x}$	-2	-1	0	1	2	0
$y - \bar{y}$	-2	0	1	0	1	0
$(x - \bar{x})^2$	4	1	0	1	4	10
$(x - \bar{x})(y - \bar{y})$	4	0	0	0	2	6
$(y - \bar{y})^2$	4	0	1	0	1	6

$$\bar{x} = \frac{15}{5} = 3, \quad \bar{y} = \frac{20}{5} = 4, \quad S_{xx} = 10, \quad S_{xy} = 6, \quad S_{yy} = 6.$$

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{6}{10} = 0.6, \quad b_0 = \bar{y} - b_1\bar{x} = 4 - 0.6 \times 3 = 2.2.$$

$$\hat{y} = 2.2 + 0.6x$$

An extra hour of revision is associated with 0.6 more marks. Check against the raw-sum forms: $S_{xx} = \sum x_i^2 - n\bar{x}^2 = 55 - 5(9) = 10$ and $S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 66 - 5(3)(4) = 6$. Same numbers.

```
n = 5
xbar = 3.0000  ybar = 4.0000
Sxx = 10.0000  Sxy = 6.0000  Syy = 6.0000
b1 = Sxy/Sxx = 0.6000
b0 = ybar - b1*xbar = 2.2000
fitted line: yhat = 2.2 + 0.6 x
```

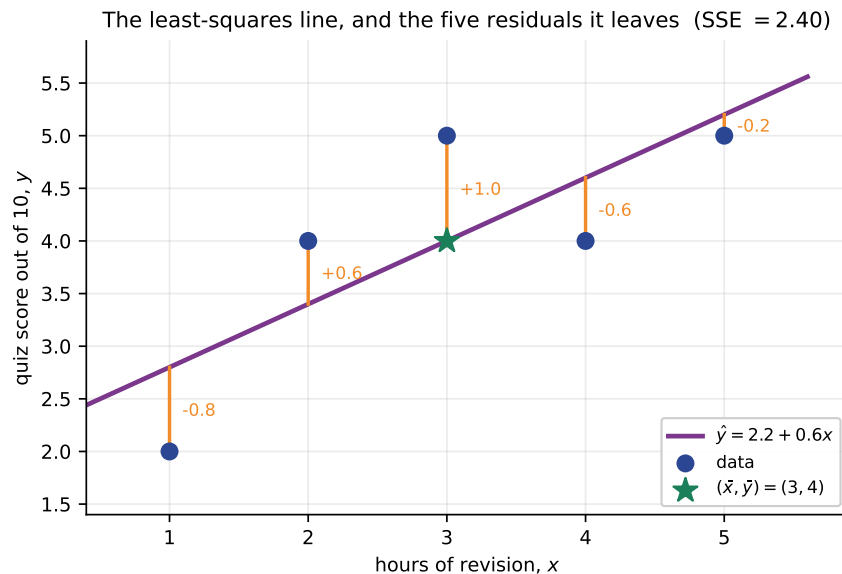


Figure 5: The least-squares line and the five residuals. They sum to exactly zero, and the green star at $(\bar{x}, \bar{y}) = (3, 4)$ lies on the line. Both facts are forced by the normal equations, as Section 9 shows.

8.2 And in code

```
x = np.array([1.0, 2.0, 3.0, 4.0, 5.0])
y = np.array([2.0, 4.0, 5.0, 4.0, 5.0])
xbar, ybar = x.mean(), y.mean()
Sxx = np.sum((x - xbar)**2)
Sxy = np.sum((x - xbar) * (y - ybar))
b1 = Sxy / Sxx
b0 = ybar - b1 * xbar
```

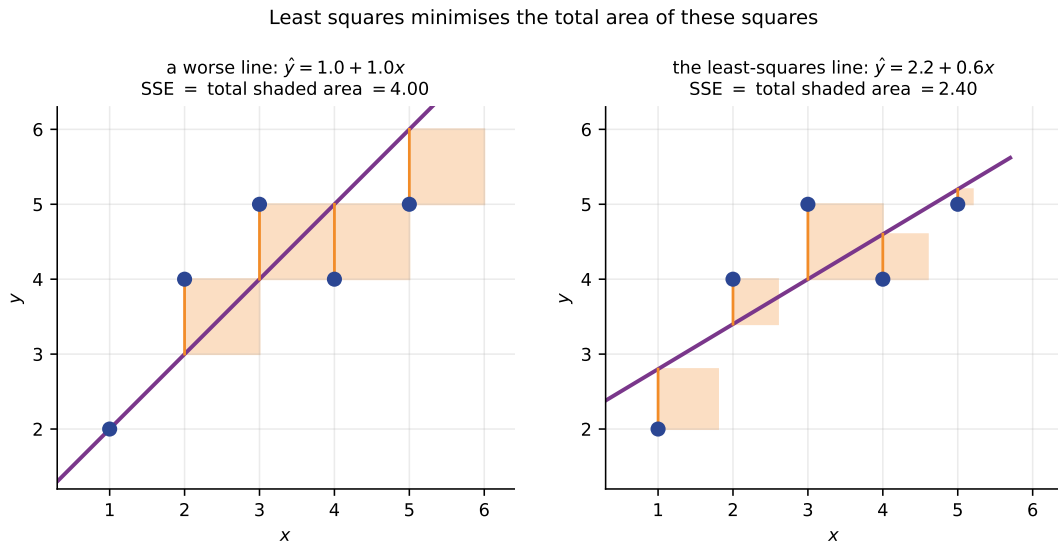


Figure 6: Each residual drawn as a literal square of side $|e_i|$. The SSE is the total shaded area: 4.00 for $\hat{y} = 1.0 + 1.0x$ and 2.40 for the least-squares line. Least squares slides the line until that area is as small as it can be.

```
p = np.polyfit(x, y, 1) # returns [slope, intercept]
lr = LinearRegression().fit(x.reshape(-1, 1), y)
```

```
by hand          b0 = 2.200000000000    b1 = 0.600000000000
np.polyfit       b0 = 2.200000000000    b1 = 0.600000000000
LinearRegression b0 = 2.200000000000    b1 = 0.600000000000
max abs difference, hand vs polyfit      4.441e-16
max abs difference, hand vs LinearRegression 0.000e+00
```

Not “close”: `LinearRegression` matched the hand arithmetic **exactly**, and `np.polyfit` differs by one unit in the last place of a double because it routes through a singular value decomposition (SVD) — a factorisation of X itself that solves least squares without ever forming $X^T X$, which is numerically safer.

Key idea

The library is not doing anything you did not just do on paper. It is doing it faster, and with better numerics on problems where $X^T X$ is ill-conditioned. If you can compute $b_1 = S_{xy}/S_{xx}$ by hand, you know what `fit()` does.

8.3 Four candidate lines

Before accepting the formula, check that the line it produces really is the best of the alternatives, and check something subtler at the same time.

Candidate	SSE	$\sum_i e_i$	$\sum_i x_i e_i$
A: $\hat{y} = 2.0 + 0.7x$	2.5500	-0.5000	-2.5000
B: $\hat{y} = 2.2 + 0.6x$	2.4000	0.0000	0.0000
C: $\hat{y} = 2.5 + 0.5x$	2.5000	0.0000	+1.0000
D: $\hat{y} = 1.0 + 1.0x$	4.0000	0.0000	-4.0000

$\sum e_i = 0$ is necessary, not sufficient

Three of the four lines have residuals summing to zero, and two of those three are worse than the least-squares line. Zero residual sum is only *one* of the two normal equations. B is the only candidate that also satisfies $\sum_i x_i e_i = 0$, and B is the one with the smallest SSE. A student who checks only the first condition has checked half the answer.

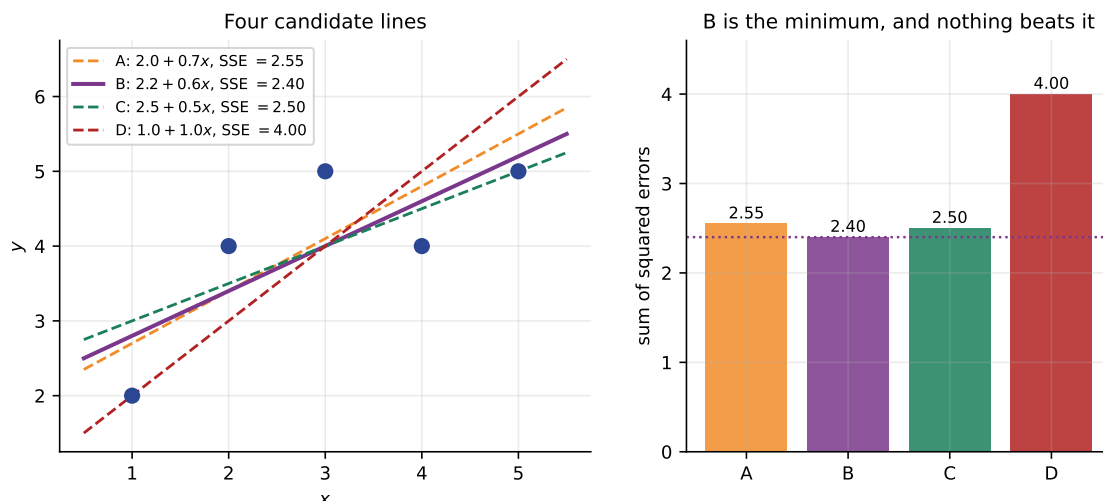


Figure 7: The four candidates and their scores. C looks almost as good as B and scores 2.50 against 2.40: near the bottom of a bowl the surface is flat, which is also why gradient descent takes so long to finish (Section 11).

9 The line of best fit and its properties

Four properties follow immediately from the two normal equations. They are short proofs and they are examined.

9.1 Property 1: the residuals sum to zero

Equation (6) was $\sum_i (y_i - b_0 - b_1 x_i) = 0$, and the bracket is e_i . Hence

$$\sum_{i=1}^n e_i = 0, \quad \text{and dividing by } n, \quad \bar{\hat{y}} = \bar{y}. \quad (17)$$

The fitted values have the same mean as the observations.

Only with an intercept

Equation (17) is the $\partial \text{SSE} / \partial b_0 = 0$ condition. Remove the intercept from the model and there is no such equation, and no such guarantee. See Section 9.6, where the residuals of a no-intercept fit sum to +2.

9.2 Property 2: the line passes through the centroid

From (8), $b_0 = \bar{y} - b_1 \bar{x}$, so evaluating the fitted line at $x = \bar{x}$:

$$\hat{y}(\bar{x}) = b_0 + b_1 \bar{x} = (\bar{y} - b_1 \bar{x}) + b_1 \bar{x} = \bar{y}. \quad (18)$$

Every least-squares line with an intercept passes through $(\bar{x}, \bar{y})$, the **centroid** (the point of means) of the data. This is a good sanity check on any hand computation: if your line misses the centroid, you have made an arithmetic error.

9.3 Property 3: residuals are orthogonal to the predictor

Equation (7) says $\sum_i x_i(y_i - b_0 - b_1 x_i) = 0$, that is

$$\sum_{i=1}^n x_i e_i = 0. \quad (19)$$

Combining with (17),

$$\sum_i \hat{y}_i e_i = \sum_i (b_0 + b_1 x_i) e_i = b_0 \underbrace{\sum_i e_i}_0 + b_1 \underbrace{\sum_i x_i e_i}_0 = 0. \quad (20)$$

Two vectors are **orthogonal** when their dot product is zero, which is exactly what (19) and (20) say. Geometrically the residual vector is orthogonal both to the column of ones and to the column x , hence to every vector in the plane they span — which is to say, **there is no linear signal left in the residuals**. If you regressed e on x you would get a slope of exactly zero.

9.4 Property 4: the sum of squares decomposes

Write $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + e_i$ and square:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i e_i^2 + 2 \sum_i (\hat{y}_i - \bar{y}) e_i.$$

The cross term is $2 \sum_i \hat{y}_i e_i - 2\bar{y} \sum_i e_i = 0$ by (20) and (17), so

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i e_i^2}_{\text{SSE}}. \quad (21)$$

Total variation splits into the part the line explains and the part it does not. One further identity, obtained by substituting $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$:

$$\text{SSR} = b_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}. \quad (22)$$

9.5 All four, on the five points

```

x      y      yhat  residual
1      2.0     2.80   -0.80
2      4.0     3.40    0.60
3      5.0     4.00    1.00
4      4.0     4.60   -0.60
5      5.0     5.20   -0.20
sum of residuals      -0.0000000000000000
sum of x_i * e_i      -0.0000000000000000
sum of yhat_i * e_i    -0.0000000000000001
mean of yhat = 4.0000   and ybar = 4.0000
the line at xbar = 3.0 gives 4.0000   and ybar = 4.0000

SSE (residual) = 2.4000
SSR (explained) = 3.6000
SST (total) = 6.0000
SSR + SSE = 6.0000   and SST = 6.0000
check: SSR = Sxy^2/Sxx = 3.6000
mean squared error of the fit = SSE/n = 0.4800
correlation r = Sxy/sqrt(Sxx*Syy) = 0.7746
and b1 = r * sd(y)/sd(x) = 0.7746 * 1.2247/1.5811 = 0.6000
r^2 = 0.6000, and SSR/SST = 0.6000 -- the same number (Lecture 9 names it)

```

$SSR = S_{xy}^2 / S_{xx} = 36/10 = 3.6$ exactly as (22) predicts, and (11) reproduces the slope from the correlation. The last line is a forward reference: $SSR/SST = 0.6$ and $r^2 = 0.6$ are the same number, and Lecture 9 gives it a name and a warning.

9.6 Drop the intercept and watch Property 1 die

Fit $\hat{y} = b_1x$ instead. There is now one parameter, hence one normal equation — the one from $\partial/\partial b_1$ — and minimising $\sum_i (y_i - b_1x_i)^2$ gives $-2\sum_i x_i(y_i - b_1x_i) = 0$, so $b_1 = \sum_i x_i y_i / \sum_i x_i^2$.

```
b1_noint = np.sum(x * y) / np.sum(x * x) # fit y = b1 x, no intercept
```

```
no-intercept fit  yhat = 1.2000 x
residuals        [ 0.8  1.6  1.4 -0.8 -1. ]
sum of residuals +2.0000 (NOT zero)
sum of x_i * e_i +1.7764e-15 (still zero -- that equation was solved)
SSE              6.8000 against 2.4000 with an intercept
```

The residuals now sum to +2, and the SSE nearly trebles. Note what *did* survive: $\sum x_i e_i$ is still zero, because that equation was the one we solved.

Key idea

“The residuals of a regression sum to zero” is a statement about the model containing an intercept, not about least squares. Whenever you see the claim, the words “with an intercept” are doing all the work.

10 How precise is the line? S_{xx} is the answer

The estimates b_0 and b_1 are computed from data, and data is noisy, so they are random quantities: collect a fresh sample, refit, and you get slightly different numbers. The distribution of b_1 over all such repeated samples is its **sampling distribution**; its standard deviation, $\sqrt{\text{Var}(b_1)}$, is the **standard error** of b_1 . Under the four assumptions of Section 5 one can show

$$\mathbb{E}[b_1] = \beta_1, \quad \text{Var}(b_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{Var}(b_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right). \quad (23)$$

(The first says least squares is *unbiased*: on average over repeated samples it hits the true value; the Gauss–Markov theorem strengthens this to “no other linear unbiased estimator has smaller variance”. Lecture 9 does the derivations properly.)

Read the middle formula. S_{xx} sits in the denominator, so the more spread out the x values, the more precisely you learn the slope — with the same number of rows, at no extra cost.

A design lesson worth money

If you can *choose* where to observe — a calibration, an experiment, a load test — put your observations at the extremes of the usable range rather than clustering them in the middle. Same n , smaller standard error.

Measure it: 4000 refits at four spreads

```
rng = np.random.default_rng(0) # the one seed for this lecture
for spread in (0.5, 1.0, 2.0, 4.0):
    xs = np.linspace(-spread, spread, 20)
    Sxx_s = np.sum((xs - xs.mean())**2)
    slopes = [np.sum((xs - xs.mean())*(ys - ys.mean())) / Sxx_s
               for ys in (1.0 + 2.0*xs + rng.normal(0, 1, 20)
                           for _ in range(4000))]
```

half-range of x	Sxx	sd of 4000 fitted slopes	sigma/sqrt(Sxx)
0.5	1.842	0.7237	0.7368
1.0	7.368	0.3683	0.3684
2.0	29.474	0.1859	0.1842
4.0	117.895	0.0932	0.0921

Eight times the range, 64 times S_{xx} , and 7.8 times the precision. The measured column and the theoretical column agree to three decimal places, so (23) is not decoration.

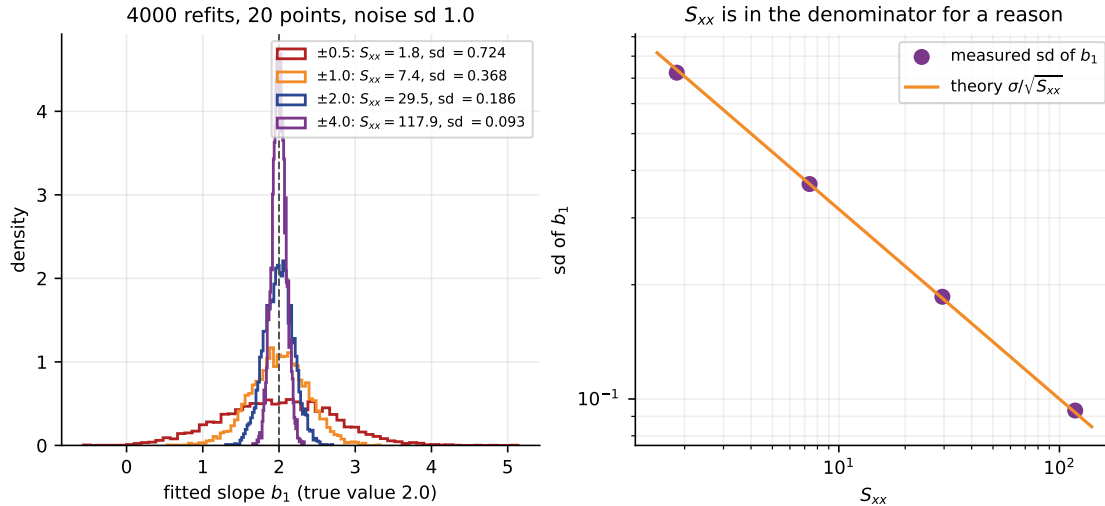


Figure 8: Left: four sampling distributions of b_1 , all centred on the true value 2.0 — least squares is unbiased whatever the spread — with very different widths. Right: the measured widths against $\sigma/\sqrt{S_{xx}}$ on logarithmic axes.

11 Two roads to the same answer

11.1 The connection to Lecture 3

Lecture 3 minimised, for a linear model,

$$J(w) = \frac{1}{n} \|Xw - y\|^2, \quad \nabla J(w) = \frac{2}{n} X^\top (Xw - y), \quad w \leftarrow w - \eta \nabla J(w),$$

and iterated until the parameters stopped moving. Compare with (15): setting that same gradient to zero gives $X^\top Xw = X^\top y$, which is a linear system you solve.

Key idea

Gradient descent *approaches* the point where the gradient vanishes. Least squares *solves* for it. They are not competing methods for different problems — for this problem they are two routes to the same point. Lecture 3 iterated because in general $\nabla J = 0$ cannot be solved in closed form; for a linear model with squared loss, it can.

11.2 On the five points

```
def grad(theta):
    r = y - (theta[0] + theta[1] * x)
    return np.array([-2.0 * r.sum(), -2.0 * np.sum(x * r)])

theta = np.array([0.0, 0.0])
for k in range(1, 4001):
    theta = theta - 0.01 * grad(theta)
```

iter	b0	b1	SSE	max theta-theta*
1	0.4000	1.3200	8.2320	1.80e+00
2	0.3640	1.0680	5.5234	1.84e+00
5	0.4655	1.0807	5.1381	1.73e+00
10	0.6072	1.0412	4.7089	1.59e+00
50	1.3946	0.8231	2.9903	8.05e-01
200	2.1376	0.6173	2.4035	6.24e-02
1000	2.2000	0.6000	2.4000	7.44e-08
4000	2.2000	0.6000	2.4000	1.24e-14
closed form		2.2000	0.6000	2.4000
gradient descent matched the closed form to 4 decimals at iteration 619				

Six hundred and nineteen iterations to reproduce one division. And look at the SSE column: after 200 steps the loss is 2.4035 against the optimal 2.4000, while the parameters are still 0.06 away. **Near the bottom of a bowl the surface is flat, so a large parameter error costs almost no loss** — which is exactly why choosing η and a stopping rule felt so fiddly in Lecture 3.

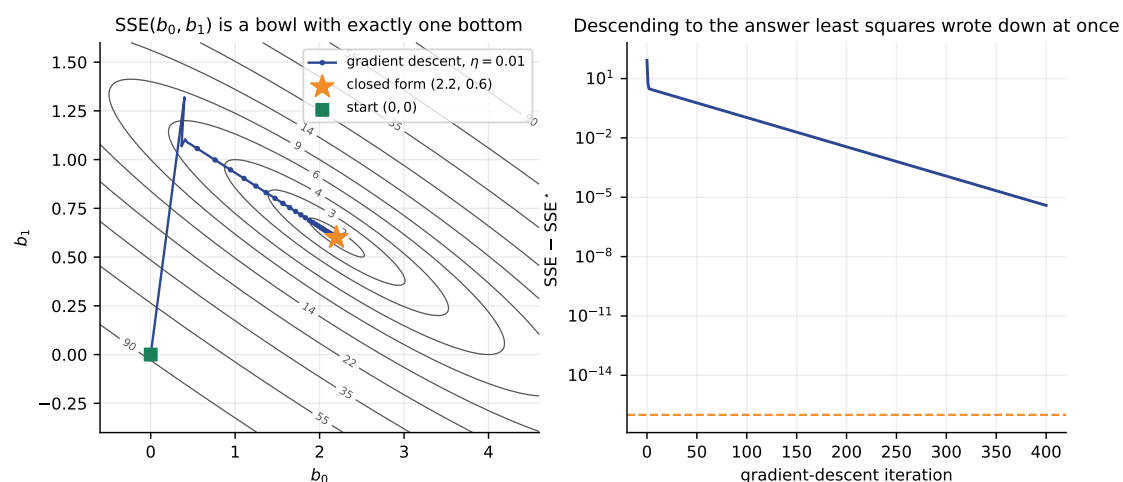


Figure 9: Left: contours of $SSE(b_0, b_1)$ with the gradient-descent path from $(0,0)$ and the closed-form solution as a star. The contours are tilted ellipses because b_0 and b_1 are correlated — the off-diagonal $\sum x_i$ of $X^T X$ — so descent zig-zags down a valley, which is the Lecture 6 ravine picture again. Right: the gap to the optimal loss on a logarithmic scale.

11.3 And on 442 real patients

The same comparison on `load_diabetes`, `bmi` against progression. Gradient descent is run on standardised x (Lecture 6's lesson: otherwise one step size does not work) and the coefficients are then unwound to the original units.

```

xs = StandardScaler().fit(xb)
Z = np.column_stack([np.ones(len(yd)), xs.transform(xb).ravel()])
wz = np.zeros(2)
for it in range(2000):
    wz = wz - 0.1 * (2.0 / len(yd)) * (Z.T @ (Z @ wz - yd))
b1_gd = wz[1] / xs.scale_[0]
b0_gd = wz[0] - wz[1] * xs.mean_[0] / xs.scale_[0]

sgd = make_pipeline(StandardScaler(), # the stochastic route
    SGDRegressor(loss="squared_error", penalty=None,
        learning_rate="constant", eta0=0.01,
        max_iter=50000, tol=1e-10,
        random_state=0)) # the one seed for this lecture
sgd.fit(xb, yd)

```

```

full-batch GD      1 steps  b0 =  30.4267  b1 =  189.8871  J = 20008.2328
full-batch GD     10 steps  b0 = 135.7983  b1 =  847.4904  J =  4180.8086
full-batch GD    100 steps  b0 = 152.1335  b1 =  949.4353  J =  3890.4566
full-batch GD    500 steps  b0 = 152.1335  b1 =  949.4353  J =  3890.4566
full-batch GD   2000 steps  b0 = 152.1335  b1 =  949.4353  J =  3890.4566
closed form                b0 = 152.1335  b1 =  949.4353  J =  3890.4566
difference in the slope    4.547e-13 (relative 4.441e-16)
SGDRegressor (stochastic) b0 = 149.5577  b1 =  891.1257  J =  3904.7835
the stochastic version stops 6.1% away from the slope: that is the
noise floor of Lecture 3, not a bug

```

One hundred steps of full-batch gradient descent (GD in the printout) reproduce $(X^T X)^{-1} X^T y$ to fifteen significant figures. **SGDRegressor** with a constant step size lands 6.1% away on the slope, and that is not a defect of the implementation: Lecture 3 established that a constant-step stochastic method converges to a *region* around the optimum, not to a point. Report it, do not hide it.

11.4 So which do you use?

	Closed form	Gradient descent
Cost	$O(nd^2 + d^3)$, one pass	$O(nd)$ per step, many steps
Memory	needs $X^T X$ or all of X	one mini-batch at a time
Tuning	none	step size, schedule, batch size, stopping
Answer	exact	as close as you can afford
Fails when	d large; $X^T X$ singular	almost never, slowly
Extends to	squared loss on a linear model	anything differentiable

Rule of thumb: small d (say up to a few thousand columns), data in memory, squared loss, linear model — **use the closed form**, which is what **LinearRegression** does. Enormous n or d , streaming data, or any model outside that family — descend. Logistic regression, two lectures away, has no closed form at all, which is why Lecture 3 came first.

12 What least squares will not tell you

12.1 Leverage: not all rows are equal

The slope is $b_1 = \sum_i (x_i - \bar{x})(y_i - \bar{y}) / S_{xx}$, so row i enters weighted by $(x_i - \bar{x})$. A row sitting at $x_i = \bar{x}$ has weight zero and *cannot* influence the slope, however wrong its y is. The **leverage**

$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \quad (24)$$

measures how much row i can pull the fit; the second term is the part that concerns the slope.

```

point moved      b0      b1      change in b1
none             2.2000   0.6000      +0.0000
y at x=3: 5 -> 9   3.0000   0.6000      +0.0000
y at x=5: 5 -> 9   0.6000   1.4000      +0.8000
y at x=1: 2 -> -2 -1.0000   1.4000      +0.8000
leverage (x-xbar)^2/Sxx: [0.4 0.1 0.  0.1 0.4]

```

The same +4 error in y leaves the slope completely unchanged when it happens at $x = \bar{x}$ and moves it by +0.8 — more than doubling it — when it happens at the end of the range. Two of the five rows carry 80% of the slope leverage between them.

Caution

“Is this point an outlier?” is the wrong question. A point far from the others *in y* may do no harm; a point far from the others *in x* can determine the answer on its own. Ask where a suspicious row sits in x before you ask how odd its y is.

Leverage $(x_i - \bar{x})^2/S_{xx} = [0.4 \ 0.1 \ 0. \ 0.1 \ 0.4]$: a point at $\bar{x}$ cannot change the slope

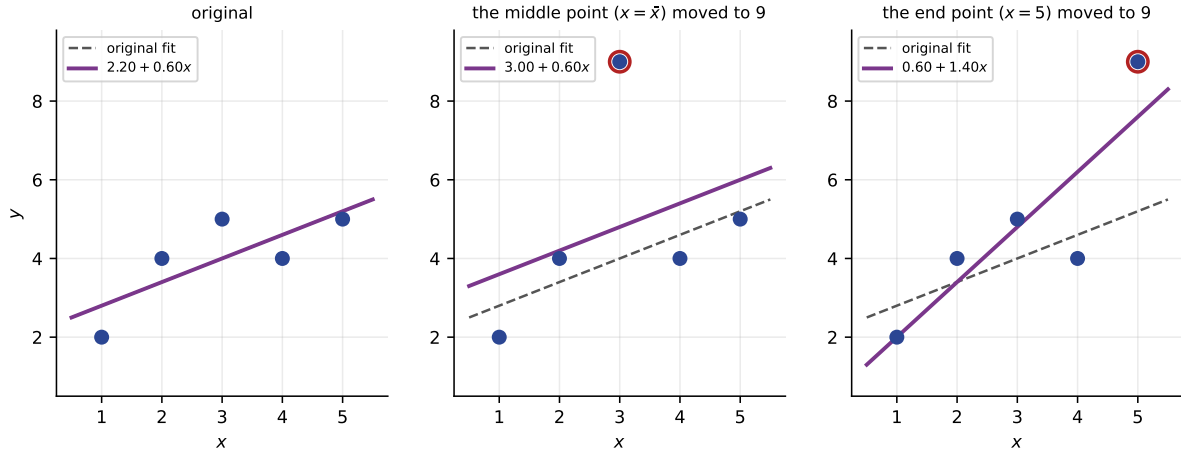


Figure 10: The same +4 change in one y value. In the middle panel the line lifts and keeps its slope; in the right panel it pivots.

12.2 Anscombe's quartet

Francis Anscombe constructed four eleven-point datasets in 1973. Every statistic least squares computes is the same for all four.

set	$\bar{x}$	$\bar{y}$	S_{xx}	S_{xy}	b_0	b_1	SSE
I	9.00	7.50	110.00	55.01	3.0001	0.5001	13.7627
II	9.00	7.50	110.00	55.00	3.0009	0.5000	13.7763
III	9.00	7.50	110.00	54.97	3.0025	0.4997	13.7562
IV	9.00	7.50	110.00	54.99	3.0017	0.4999	13.7425

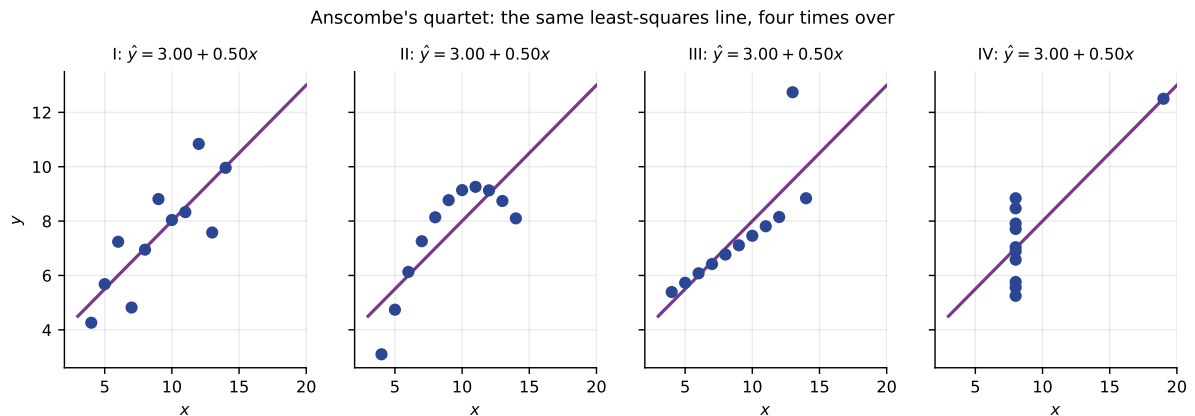


Figure 11: Anscombe's quartet. Every column of the table above agrees to two decimal places; the pictures do not agree at all.

Dataset I is a reasonable linear fit. Dataset II is a *parabola* — the model is simply wrong, and no amount of extra data would fix it. Dataset III is a perfect straight line plus one contaminated point. Dataset IV has ten observations at $x = 8$ and a single observation at $x = 19$ whose leverage is 1.0: that one point determines the slope, and deleting it leaves $S_{xx} = 0$ and no fit at all.

Key idea

Least squares always returns a line. Whether the line means anything is a separate question, and the estimates alone cannot answer it. Plot the data, and plot the residuals — which is Lecture 9's business.

12.3 And it says nothing about causation

feature	b1	b0	SSE	corr(x,y)
bmi	949.435	152.133	1719581.8	0.5865
s5	916.137	152.133	1781701.4	0.5659
bp	714.738	152.133	2110158.3	0.4415
s4	696.883	152.133	2135363.2	0.4305
s3	-639.145	152.133	2212502.4	-0.3948
s6	619.223	152.133	2237572.2	0.3825
s1	343.254	152.133	2503185.5	0.2120
age	304.183	152.133	2528481.8	0.1879
s2	281.785	152.133	2541606.6	0.1741
sex	69.715	152.133	2616148.9	0.0431
total sum of squares about ybar:			2621009.1	

Every intercept is 152.133, because $b_0 = \bar{y} - b_1\bar{x}$ and every column of this dataset has already been mean-centred, so $\bar{x} = 0$ and $b_0 = \bar{y}$ regardless of the slope. That is a good check that you have understood (8).

The best single predictor removes $1 - 1719581.8/2621009.1 = 34\%$ of the total sum of squares; the worst removes 0.2%. None of this is causation: **bmi** predicting progression is a statement about the joint distribution in this table, not about what happens to a patient whose BMI is changed.

13 Summary

1. Regression models a quantitative response as $y = f(x) + \varepsilon$. It is done either for prediction or for explanation, and the two are judged differently.
2. A model is **linear** when it is linear in the *parameters*. A parabola is a linear model; $\beta_0 + \beta_0\beta_1x$ is not. Linear models are fitted by solving one linear system; everything else needs an iterative solver.
3. Eight steps: problem, variables, data, specification, fitting method, fit, validation, use. Steps 1–3 decide the analysis; step 7 sends you back to step 4, not step 6.
4. Least squares minimises $\text{SSE}(b_0, b_1) = \sum_i (y_i - b_0 - b_1x_i)^2$. Squares are chosen because the criterion is differentiable everywhere and its derivative is *linear* in the unknowns.
5. Setting both partial derivatives to zero gives the **normal equations** $nb_0 + b_1 \sum x_i = \sum y_i$ and $b_0 \sum x_i + b_1 \sum x_i^2 = \sum x_i y_i$, whose solution is $b_1 = S_{xy}/S_{xx}$ and $b_0 = \bar{y} - b_1\bar{x}$.
6. The Hessian is $2(\sum_{x_i}^n \sum x_i^2)$ with $\det = 4nS_{xx} > 0$: strictly convex, so the stationary point is the unique global minimum. $S_{xx} = 0$ is the one failure mode.
7. On five points with $S_{xx} = 10$, $S_{xy} = 6$: $\hat{y} = 2.2 + 0.6x$, $\text{SSE} = 2.40$. `np.polyfit` agreed to 4×10^{-16} and `LinearRegression` to 0.
8. In matrix form $X^\top X b = X^\top y$, and by hand $\frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} \begin{pmatrix} 20 \\ 66 \end{pmatrix} = (2.2, 0.6)$. The formula never mentions the number of columns.

9. With an intercept, $\sum e_i = 0$, the line passes through $(\bar{x}, \bar{y})$, $\sum x_i e_i = 0$ and $\text{SST} = \text{SSR} + \text{SSE}$. Drop the intercept and the first of those became $+2.0$. Zero residual sum is necessary but **not sufficient**: three of four candidate lines had it.
10. Gradient descent on the identical objective took **619 iterations** to match one division on the five points, and on diabetes reproduced the closed form to a relative 4×10^{-16} . Constant-step stochastic descent stopped 6.1% short — the Lecture 3 noise floor.
11. $\text{Var}(b_1) = \sigma^2/S_{xx}$: widening the range of x eightfold cut the spread of the fitted slope by $7.8\times$ on the same 20 rows. Spread your x values out if you can choose them.
12. Leverage $(x_i - \bar{x})^2/S_{xx}$ decides which rows matter: a point at $\bar{x}$ cannot change the slope at all. **Anscombe's quartet** has four datasets sharing every computed statistic and four completely different pictures.

14 Practice questions

Q1. (By hand.) For

$$x = 2, 4, 6, 8, 10, \quad y = 3, 7, 5, 10, 12,$$

compute $\bar{x}$, $\bar{y}$, S_{xx} , S_{xy} and S_{yy} ; give the least-squares line; list the five residuals and verify they sum to zero; compute SSE, SSR and SST and check (21); and predict y at $x = 7$.

Q2. (By hand, matrix route.) For $x = 0, 1, 2, 3$ and $y = 1, 3, 2, 5$, write down the design matrix X , form $X^\top X$ and $X^\top y$, compute $\det(X^\top X)$ and $(X^\top X)^{-1}$ exactly, and obtain b from (16). Confirm your slope and intercept against S_{xy}/S_{xx} and $\bar{y} - b_1\bar{x}$.

Q3. Prove, from the normal equations only, that a least-squares line with an intercept satisfies $\sum_i e_i = 0$ and passes through $(\bar{x}, \bar{y})$. Then give a one-line reason why a no-intercept fit need not satisfy the first of these.

Q4. Write down the Hessian of $\text{SSE}(b_0, b_1)$, show that its determinant equals $4nS_{xx}$, and state the exact condition under which the least-squares solution exists and is unique. Give a five-row dataset for which it does not.

Q5. Four lines are proposed for the data $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$: A $\hat{y} = 2.0 + 0.7x$, B $\hat{y} = 2.2 + 0.6x$, C $\hat{y} = 2.5 + 0.5x$, D $\hat{y} = 1.0 + 1.0x$. Three of them have residuals summing to zero. Without computing any SSE, explain how to decide which is the least-squares line, and carry out your test.

Q6. (By hand.) Derive the least-squares estimator for the no-intercept model $y = \beta_1 x + \varepsilon$, then apply it to the data of Q1. Compare its residual sum and its SSE with the answers from Q1 and say why they differ.

Q7. Derive the least-squares estimator for simple linear regression from scratch on a blank page: state the criterion, take both partial derivatives, set them to zero, obtain and solve the normal equations, and verify the second-order condition. This is the derivation the examination will ask for.

Q8. For the design of Q1, compute the leverage $(x_i - \bar{x})^2/S_{xx}$ of each of the five points. Then predict, *before* computing, what happens to b_1 if you add 5 to each y_i in turn, and check your predictions.

Q9. You may take ten observations with x anywhere in $[0, 10]$ and you want the most precise possible estimate of the slope. Compute S_{xx} for four designs — ten equally spaced points;

five at $x = 0$ and five at $x = 10$; ten points equally spaced in $[4, 6]$; all ten at $x = 5$ — and rank them. What practical reason might make you reject the best one?

- Q10.** A classmate fits the same simple linear regression twice. With `LinearRegression` the slope is 949.44; with `SGDRegressor(learning_rate="constant", eta0=0.01)` it is 891.13, a difference of 6.1%. They ask whether there is a bug. Answer them, and describe two changes that would bring the second number closer to the first.

15 Solutions

- Q1.** $\bar{x} = 30/5 = 6$ and $\bar{y} = 37/5 = 7.4$. Build the deviation table:

						Σ
$x - \bar{x}$	-4	-2	0	2	4	0
$y - \bar{y}$	-4.4	-0.4	-2.4	2.6	4.6	0
$(x - \bar{x})^2$	16	4	0	4	16	40
$(x - \bar{x})(y - \bar{y})$	17.6	0.8	0	5.2	18.4	42
$(y - \bar{y})^2$	19.36	0.16	5.76	6.76	21.16	53.2

So $b_1 = 42/40 = 1.05$ and $b_0 = 7.4 - 1.05 \times 6 = 7.4 - 6.3 = 1.1$, and $\hat{y} = 1.1 + 1.05x$. Fitted values 3.2, 5.3, 7.4, 9.5, 11.6; residuals -0.2, 1.7, -2.4, 0.5, 0.4, which sum to 0 exactly. $SSE = 0.04 + 2.89 + 5.76 + 0.25 + 0.16 = 9.10$; $SSR = S_{xy}^2/S_{xx} = 1764/40 = 44.10$; $SST = 53.20$, and $44.10 + 9.10 = 53.20$. Prediction at $x = 7$: $1.1 + 1.05 \times 7 = 8.45$.

```
xbar 6.00 ybar 7.40 Sxx 40.00 Sxy 42.00 Syy 53.20
b1 1.0500 b0 1.1000 residuals [-0.2 1.7 -2.4 0.5 0.4] sum 4.4e-16
SSE 9.10 SSR 44.10 SST 53.20 polyfit [1.1 1.05]
prediction at x = 7: 8.450
```

Note the third point: $\hat{y}_3 = 7.4 = \bar{y}$, because $x_3 = \bar{x}$ and the line passes through the centroid.

- Q2.** $n = 4$, $\sum x_i = 6$, $\sum x_i^2 = 0 + 1 + 4 + 9 = 14$, $\sum y_i = 11$, $\sum x_i y_i = 0 + 3 + 4 + 15 = 22$. Hence

$$X^T X = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, \quad X^T y = \begin{pmatrix} 11 \\ 22 \end{pmatrix}, \quad \det = 56 - 36 = 20, \quad (X^T X)^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}.$$

$$b = \frac{1}{20} \begin{pmatrix} 14 \cdot 11 - 6 \cdot 22 \\ -6 \cdot 11 + 4 \cdot 22 \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 154 - 132 \\ -66 + 88 \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 22 \\ 22 \end{pmatrix} = \begin{pmatrix} 1.1 \\ 1.1 \end{pmatrix}.$$

Scalar check: $\bar{x} = 1.5$, $\bar{y} = 2.75$, $S_{xx} = 5$, $S_{xy} = 5.5$, so $b_1 = 5.5/5 = 1.1$ and $b_0 = 2.75 - 1.1(1.5) = 1.1$. Also $\det(X^T X) = nS_{xx} = 4 \times 5 = 20$, as Section 7 promised.

```
X'X [[4.0, 6.0], [6.0, 14.0]] X'y [11.0, 22.0] det 20.0
inv(X'X)*20 = [[14.0, -6.0], [-6.0, 4.0]]
b = [1.1 1.1] LinearRegression [1.1 1.1]
residuals [-0.1 0.8 -1.3 0.6] sum 6.7e-16 SSE 2.7000
```

- Q3.** The first normal equation (6) is $\sum_i (y_i - b_0 - b_1 x_i) = 0$, and the bracket is exactly e_i , so $\sum_i e_i = 0$. Dividing the same equation by n gives $\bar{y} - b_0 - b_1 \bar{x} = 0$, i.e. $b_0 + b_1 \bar{x} = \bar{y}$, which says the fitted line evaluated at $x = \bar{x}$ returns $\bar{y}$; the line passes through $(\bar{x}, \bar{y})$.

The no-intercept model has a single parameter, so minimising produces a single normal equation, the one obtained by differentiating with respect to b_1 : $\sum_i x_i e_i = 0$. There is no $\partial/\partial b_0$ equation because there is no b_0 , and $\sum_i e_i = 0$ was *that* equation. Measured on the five lecture points, the no-intercept residuals sum to +2.0.

Q4. From (12), $H = 2(\sum_{x_i}^n \sum_{x_i^2}^{x_i})$, so $\det H = 4(n \sum x_i^2 - (\sum x_i)^2)$. Substituting $\sum x_i = n\bar{x}$ gives $4n(\sum x_i^2 - n\bar{x}^2) = 4nS_{xx}$. Since $H_{11} = 2n > 0$, H is positive definite if and only if $\det H > 0$, i.e. if and only if $S_{xx} > 0$, i.e. if and only if the x_i are **not all equal**. A dataset for which it fails: $x = (3, 3, 3, 3, 3)$ with any y at all. Then $S_{xx} = 0$, $\det(X^\top X) = 0$, the design matrix has rank 1, and infinitely many (b_0, b_1) achieve the same minimum SSE.

if every x is identical: Sxx = 0.0 and det(X'X) = 0.0 -> no unique fit

Q5. A line is the least-squares line if and only if it satisfies *both* normal equations, which in residual form are $\sum_i e_i = 0$ **and** $\sum_i x_i e_i = 0$. Since three of the four already pass the first test, apply the second. For candidate C, $\hat{y} = 2.5 + 0.5x$ gives residuals $-1.0, 0.5, 1.0, -0.5, 0.0$, and $\sum x_i e_i = -1 + 1 + 3 - 2 + 0 = +1 \neq 0$, so C is not it. The same computation gives -4.0 for D and -2.5 for A. Only B gives 0.

candidate line	SSE	sum e_i	sum x_i e_i
A: yhat = 2.0 + 0.7x	2.5500	-0.5000	-2.5000
B: yhat = 2.2 + 0.6x	2.4000	-0.0000	-0.0000
C: yhat = 2.5 + 0.5x	2.5000	+0.0000	+1.0000
D: yhat = 1.0 + 1.0x	4.0000	+0.0000	-4.0000

The SSE column confirms it: B has the smallest, and it is the only line passing both tests.

Q6. Minimise $g(b_1) = \sum_i (y_i - b_1 x_i)^2$. Then

$$g'(b_1) = \sum_i 2(y_i - b_1 x_i)(-x_i) = 0 \implies \sum_i x_i y_i = b_1 \sum_i x_i^2 \implies b_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

and $g''(b_1) = 2 \sum_i x_i^2 > 0$ unless every $x_i = 0$, so it is a minimum. On the Q1 data, $\sum x_i y_i = 6 + 28 + 30 + 80 + 120 = 264$ and $\sum x_i^2 = 4 + 16 + 36 + 64 + 100 = 220$, so $b_1 = 264/220 = 1.2$.

b1 = sum(xy)/sum(x^2) = 264.0/220.0 = 1.2000
residual sum 1.0000 (not zero) SSE 10.2000 against 9.1000

The residuals no longer sum to zero, and the SSE rises from 9.10 to 10.20. Both differences have the same cause: the no-intercept model is a *restriction* of the model in Q1 (it is the case $b_0 = 0$), and minimising over a smaller set cannot give a smaller minimum. Forcing the line through the origin also removes the equation that made the residuals cancel.

Q7. Write $\text{SSE}(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2$. Then

$$\frac{\partial \text{SSE}}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \quad \frac{\partial \text{SSE}}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

Setting both to zero and dividing by -2 :

$$nb_0 + b_1 \sum_i x_i = \sum_i y_i, \quad b_0 \sum_i x_i + b_1 \sum_i x_i^2 = \sum_i x_i y_i.$$

Divide the first by n to get $b_0 = \bar{y} - b_1 \bar{x}$; substitute into the second and collect:

$$b_1 \left(\sum_i x_i^2 - n\bar{x}^2 \right) = \sum_i x_i y_i - n\bar{x}\bar{y}, \quad \text{i.e.} \quad b_1 S_{xx} = S_{xy}, \quad b_1 = \frac{S_{xy}}{S_{xx}}.$$

For the second-order condition, the Hessian is $2(\sum_{x_i}^n \sum_{x_i^2}^{x_i})$ with leading entry $2n > 0$ and determinant $4nS_{xx} > 0$, hence positive definite: the stationary point is the unique global minimum. (Full detail in Section 6. Marks in an examination are usually distributed over: writing the criterion; both partial derivatives; both normal equations; the substitution; the identification of S_{xx} and S_{xy} ; and the second-order check — students routinely lose the last one by stopping at the first-order conditions.)

Q8. $\bar{x} = 6$ and $S_{xx} = 40$, so the leverages $(x_i - \bar{x})^2/S_{xx}$ are $16/40, 4/40, 0, 4/40, 16/40 = 0.4, 0.1, 0, 0.1, 0.4$. Adding 5 to y_i changes S_{xy} by $5(x_i - \bar{x})$ and hence changes b_1 by $5(x_i - \bar{x})/S_{xx}$, which is $-0.5, -0.25, 0, +0.25, +0.5$ respectively.

```
[0.4 0.1 0.  0.1 0.4]
add 5 to y1 -> b1 = 0.5500 (change -0.5000)
add 5 to y2 -> b1 = 0.8000 (change -0.2500)
add 5 to y3 -> b1 = 1.0500 (change +0.0000)
add 5 to y4 -> b1 = 1.3000 (change +0.2500)
add 5 to y5 -> b1 = 1.5500 (change +0.5000)
```

The middle point is inert: it sits at $\bar{x}$, so its weight $(x_i - \bar{x})$ is zero and only b_0 responds.

equally spaced	Sxx	101.852	sd(b1) proportional to 0.0991
5 at each end	Sxx	250.000	sd(b1) proportional to 0.0632
clustered 4 to 6	Sxx	4.074	sd(b1) proportional to 0.4954
all at 5	Sxx	0.000	sd(b1) proportional to infinite

Ranking by S_{xx} : five-at-each-end (250) beats equally spaced (101.9) beats clustered (4.07) beats degenerate (0). The two-point design gives a standard error $0.0632/0.0991 = 0.64$ times that of equal spacing — a 36% improvement for free.

The practical objection is decisive: with observations at only two values of x you **cannot detect departures from linearity**. Two points determine a line, and any curvature between them is invisible. The two-point design is optimal only if you already know the relationship is a straight line. If you do not, equal spacing buys you the ability to look — which, after Anscombe's quartet, you should want.

Q10. There is almost certainly no bug. The two methods minimise the same objective, but `LinearRegression` solves $X^T X b = X^T y$ exactly, while `SGDRegressor` takes noisy steps using one observation at a time. Lecture 3 established that with a *constant* step size a stochastic method converges to a region around the optimum rather than to the optimum itself; the width of that region is set by η and by the gradient noise. A 6% discrepancy in the slope with an essentially unchanged loss ($J = 3904.8$ against 3890.5, a difference of 0.4%) is exactly the flat-bottomed-bowl phenomenon of Section 11: near the optimum, large parameter errors cost very little loss.

Two changes that close the gap: (i) use a **decaying** step size (`learning_rate="invscaling"` or `"optimal"`, or `average=True`), so the steps shrink as the iterates approach the optimum; (ii) use **full-batch** gradient descent instead of stochastic, which removes the gradient noise entirely — in these notes 100 full-batch steps matched the closed form to a relative 4×10^{-16} . A third, always worth doing, is to standardise x first, which is what makes a single step size workable at all.

And the real answer to the classmate: for a problem this size, do not use `SGDRegressor`. Two parameters and 442 rows is exactly the regime the closed form was designed for.

16 References and further reading

All links were checked on 22 August 2026. Free resources are marked [free].

Textbooks

- James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Python*. [free] **§3.1 is the prescribed reading for this lecture** — §3.1.1 estimating the

coefficients (the same S_{xy}/S_{xx} derivation, with the same advertising example), §3.1.2 the accuracy of the estimates and the standard error formulas of Section 10, §3.1.3 R^2 and the residual standard error (Lecture 9). §3.2 is Lecture 11. <https://www.statlearning.com/>

- Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006. **The prescribed text.** §3.1 linear basis function models — §3.1.1 derives exactly the matrix normal equations of Section 7 and their maximum-likelihood reading (Lecture 9), §3.1.2 the geometry of least squares, §3.1.3 sequential (gradient) learning, which is the Lecture 3 half of Section 11. A free PDF is hosted by Microsoft Research; their site blocks automated link checks, so no direct URL is printed here.
- Deisenroth, Faisal and Ong, *Mathematics for Machine Learning*, Cambridge, 2020. **[free] §9.2 “Parameter Estimation”** is the reference for the derivation in Section 6, done carefully in the matrix notation, with the maximum-likelihood version alongside; §9.1 sets up the problem and §9.2.2 covers the closed-form solution. <https://mml-book.github.io/>
- Müller and Guido, *Introduction to Machine Learning with Python*, O’Reilly, 2016. §2.3.3 linear models — the practical scikit-learn view of `LinearRegression`, and the lead-in to ridge and lasso in Lecture 12. Notebooks **[free]**: https://github.com/amueller/introduction_to_ml_with_python
- Chatterjee and Hadi, *Regression Analysis by Example*, 5th edition, Wiley, 2012. Chapter 1 is the source of the eight-step workflow in Section 4; Chapter 2 is simple linear regression at exactly this level, with worked numerical examples of the kind the examination uses.
- Montgomery, Peck and Vining, *Introduction to Linear Regression Analysis*, 5th edition, Wiley. Chapter 2 is the standard engineering treatment: the derivation, the properties of Section 9, the variance formulas of Section 10, and a large supply of numerical exercises.
- Penn State, *STAT 501: Regression Methods*. **[free]** An entire online course at this level. Lesson 1 covers everything in these notes; Lesson 2 is Lecture 9. Worth an afternoon. <https://online.stat.psu.edu/stat501/lesson/1>

Online courses

- **NPTEL, Linear Regression Analysis and Forecasting** — Prof. Shalabh, IIT Kanpur. The single closest match on NPTEL to this lecture and to Lectures 9 and 11: simple linear regression, the least-squares derivation, properties of the estimators, and model adequacy, in the vocabulary the examination uses. **[free]** <https://nptel.ac.in/courses/111104098>
- **NPTEL, Introduction to Machine Learning** — Prof. Balaraman Ravindran, IIT Madras; the reference course for the semester, with a linear-regression week. **[free]** <https://nptel.ac.in/courses/106106139>
- **SWAYAM** — the national portal hosting NPTEL, with certification cycles. **[free]** <https://swayam.gov.in/>
- **Google Machine Learning Crash Course**, “Linear regression” — an 80-minute module covering the model, the loss, gradient descent and hyperparameters, with interactive widgets. It arrives at today’s model by the Lecture 3 route, which makes it a useful contrast. **[free]** <https://developers.google.com/machine-learning/crash-course/linear-regression>
- **Kaggle Learn, Intro to Machine Learning** — four hours, entirely in the browser, no installation. **[free]** <https://www.kaggle.com/learn/intro-to-machine-learning>

Videos

- StatQuest with Josh Starmer, *The Main Ideas of Fitting a Line to Data (The Main Ideas of Least Squares and Linear Regression.)* — nine minutes, and the best available intuition for why we minimise the sum of squares. Watch this one first. [free] <https://youtu.be/PaFPbb66DxQ>
- StatQuest with Josh Starmer, *Linear Regression, Clearly Explained!!!* — the follow-up, taking least squares through to R^2 and the F test, which is Lecture 9. [free] https://youtu.be/nk2CQITm_eo
- Khan Academy, *Proof (part 1) minimizing squared error to regression line* — the derivation of Section 6 at a whiteboard, in four parts, at a deliberate pace. If the algebra in Section 6 moved too fast, this is the remedy. [free] <https://youtu.be/mIx20j5y9Q8>
- Mahesh Huddar, *Linear Regression Algorithm – Solved Numerical Example in Machine Learning* — a numerical example worked in the format an Indian university examination uses. [free] <https://youtu.be/QcPycBZomac>
- Mahesh Huddar, *Linear Regression Matrix Form | Solved Numerical Example using Matrix Method* — the same problem through $(X^T X)^{-1} X^T y$, matching Section 7. [free] <https://youtu.be/JUQXyQ2U1b8>

Documentation and interactive explainers

- Victor Powell and Lewis Lehe, *Ordinary Least Squares Regression explained visually* — drag the points and watch the line and the residual squares move. Five minutes here makes Figure 6 obvious. [free] <https://setosa.io/ev/ordinary-least-squares-regression/>
- *Seeing Theory*, Chapter 6: Regression Analysis (Brown University). [free] Uses exactly the S_{xy}/S_{xx} notation of these notes, lets you drag points in each of **Anscombe’s four datasets** and watch the OLS (ordinary least squares) table update live, and continues into correlation and ANOVA (analysis of variance). <https://seeing-theory.brown.edu/regression-analysis/index.html>
- NIST/SEMATECH, *e-Handbook of Statistical Methods*, §4.1.4.1 “Linear Least Squares Regression”. [free] The authoritative free statement of the “linear in the parameters” definition of Section 3.2, with the advantages and disadvantages of least squares set out plainly. <https://www.itl.nist.gov/div898/handbook/pmd/section1/pmd141.htm>
- scikit-learn user guide, *Linear Models* (§1.1) — §1.1.1 ordinary least squares, including the statement that it solves $\min_w \|Xw - y\|_2^2$ and the note on collinearity; §1.1.13 stochastic gradient descent as an alternative. [free] https://scikit-learn.org/stable/modules/linear_model.html
- NumPy, `Polynomial.fit` — the modern replacement for `np.polyfit`, and its documentation explains that both are least-squares fits routed through `numpy.linalg.lstsq`. [free] <https://numpy.org/doc/stable/reference/generated/numpy.polynomial.polynomial.Polynomial.fit.html>

Articles

- Anscombe, F. J., “Graphs in Statistical Analysis”, *The American Statistician* 27(1), 1973, pp. 17–21. Four pages, and the source of the quartet in Section 12.2. Every argument for plotting your data is in it.

- Stigler, S. M., “Gauss and the Invention of Least Squares”, *The Annals of Statistics* 9(3), 1981, pp. 465–474. Who invented least squares, and the Gauss–Legendre priority dispute — readable, and a reminder that the method is over two hundred years old.

Sources: James et al., *An Introduction to Statistical Learning with Python*, Ch. 3–6.; Bishop, *Pattern Recognition and Machine Learning*, Ch. 3.; Deisenroth, Faisal and Ong, *Mathematics for Machine Learning*, Ch. 9.; Müller and Guido, *Introduction to Machine Learning with Python*, §2.3.